Two challenges, one laser

First we entangled two atoms. Then we made five atoms solve a puzzle.
Checked on Pasqal's cloud — and run twice on the real quantum computer.

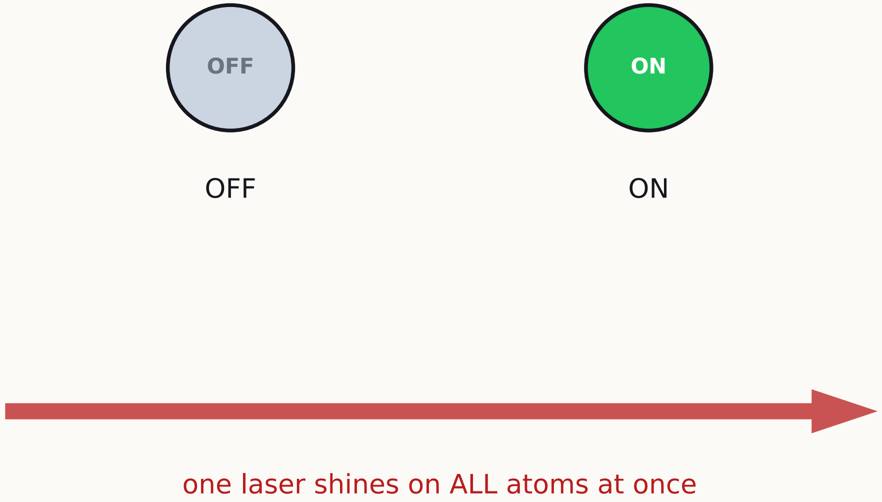
Challenge 1 score	Challenge 2 score
0.75 → 1.000000	**0.66 → 0.999999**

Team 6 · A Real Quantum Hackathon · Microsoft Garage NYC

HOW IT WORKS

The machine, in thirty seconds

Pasqal's computer holds single atoms in place with laser 'tweezers', then drives them with one control laser.



- Each atom is a switch: OFF (its normal state) or ON (a high-energy state).
- One laser drives every atom together. We choose its power and its pitch, as functions of time.
- Two ON atoms push on each other. If they are close, that push is so big that both being ON is impossible.

Six terms — this is all the jargon we need

Ω(t) 'omega'

laser POWER over time. The machine allows up to 12.6 rad/µs (from Pasqal's published spec).

δ(t) 'delta'

laser PITCH (frequency offset) over time — negative discourages ON, positive encourages it.

Blockade, R_b

the no-two-ON rule. Its reach R_b ≈ 7.2 µm follows from the machine's spec: $R_b = (C_6/\Omega)^{(1/6)}$.

Fidelity F

Challenge 1's score, defined by the organizers: $F = |\langle \text{target}|\text{our state}\rangle|^2$. 1.0 = perfect match.

MIS · P_MIS

Maximum Independent Set: biggest group with no two neighbors. P_MIS (Challenge 2's score) = the probability one measurement shows a best answer.

QPU · shot

QPU = the real quantum computer (FRESNEL, in France). A shot = one run + one photo of the atoms. All scores use 500 shots — the count fixed in the challenge brief.

Challenge 1 — make two atoms share one excitation

GIVEN by the organizers:

- Two atoms, both OFF, at spacing 5.0 µm — and then again at 6.5 µm.
- A starter pulse to beat. Its scores: F = 0.9926 (close pair) and F = 0.7500 (far pair).
- The target: the Bell state $(|gr\rangle+|rg\rangle)/\sqrt{2}$ — one atom ON, but genuinely shared between both.

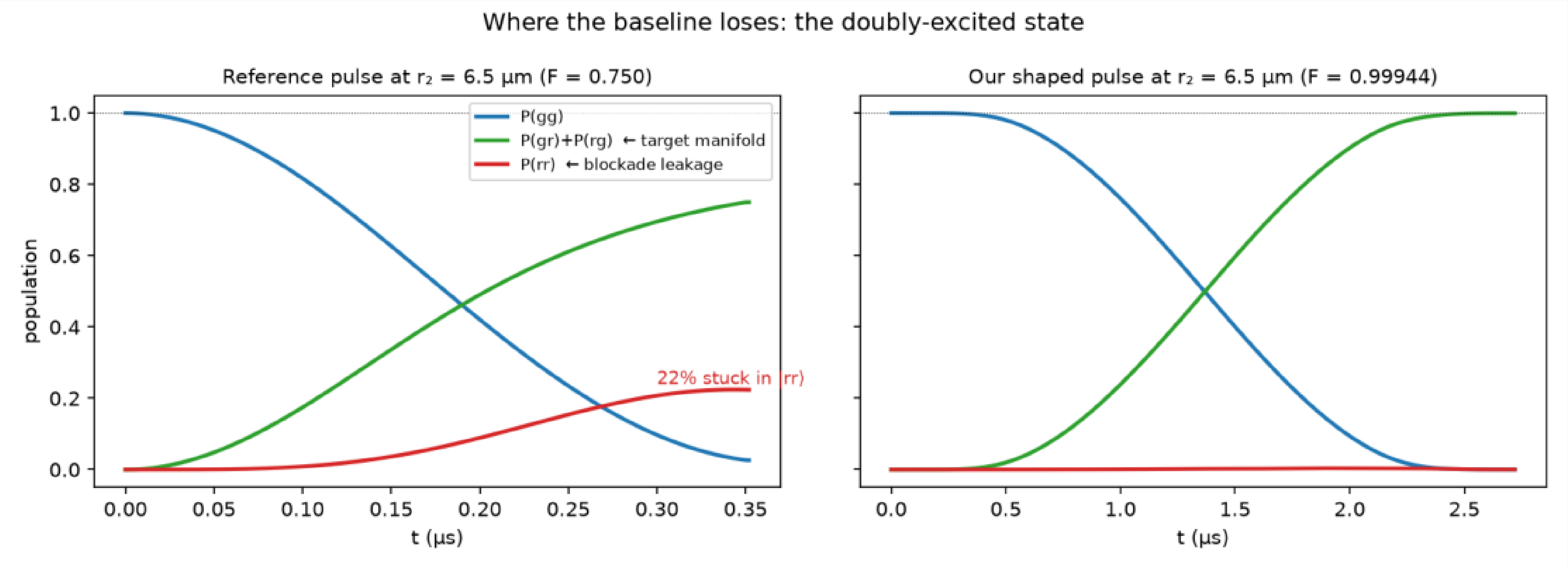
WE design:

- The laser power $\Omega(t)$ and pitch $\delta(t)$, inside the machine's published limits, at both spacings.

The whole challenge in one sentence: the starter pulse loses 25% at the far spacing — find pulses that do not.

Where the missing 25% goes

We computed the state at every moment of the pulse (exact simulation). Red = the forbidden both-ON state.



x-axis: time during the pulse, in microseconds · y-axis: probability of each two-atom outcome

- Left, the starter pulse at 6.5 µm: the atoms are far apart, the blockade is weak, and 22% leaks into the forbidden both-ON state (red). The target (green) gets stuck at 0.75.
- Right, our pulse: gentler power restores the blockade, so red stays at zero and green reaches 1.0.

How we solved it, step by step

1

Build the referee

one program that scores any pulse exactly the way the judges do

→

2

Reproduce the baseline

0.9926 and 0.7500 — now 'beat it' is a testable claim

→

3

Diagnose with physics

one ratio (push ÷ power) explains the failure; lower the power to fix it

→

4

Optimize for speed

a standard method (GRAPE, from NMR research) shapes the pulse near the physical speed limit

→

5

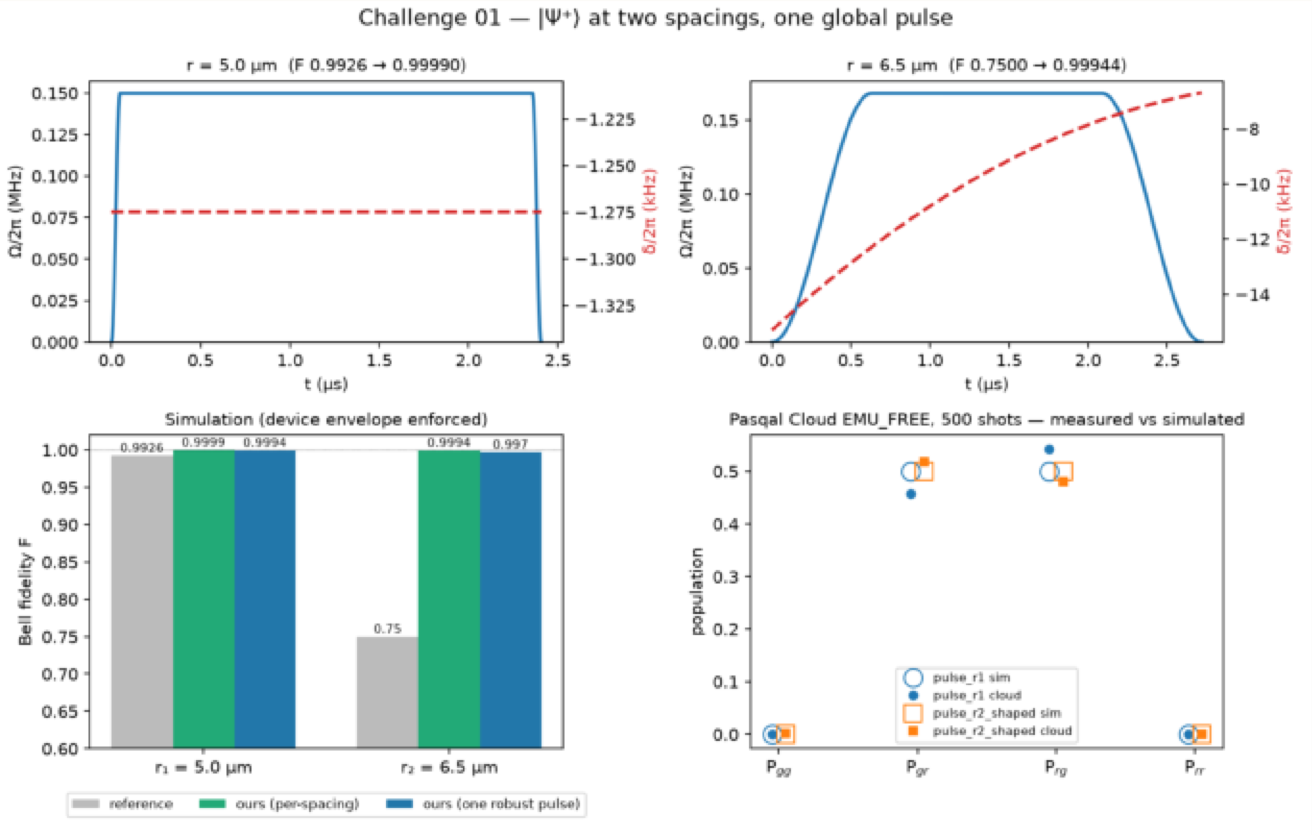
Validate outward

device rule-check → cloud, 500 shots → the real machine, 500 shots

- Each step is checked before the next: the referee has self-tests, the optimizer's math is verified against brute force, and we publish our predicted hardware numbers BEFORE each real run.
- Result: fidelity 1.000000 at both spacings, with pulses 224–600 nanoseconds long — up to 11× shorter than the starter pulse, which matters because noise grows with time.

Challenge 1 results

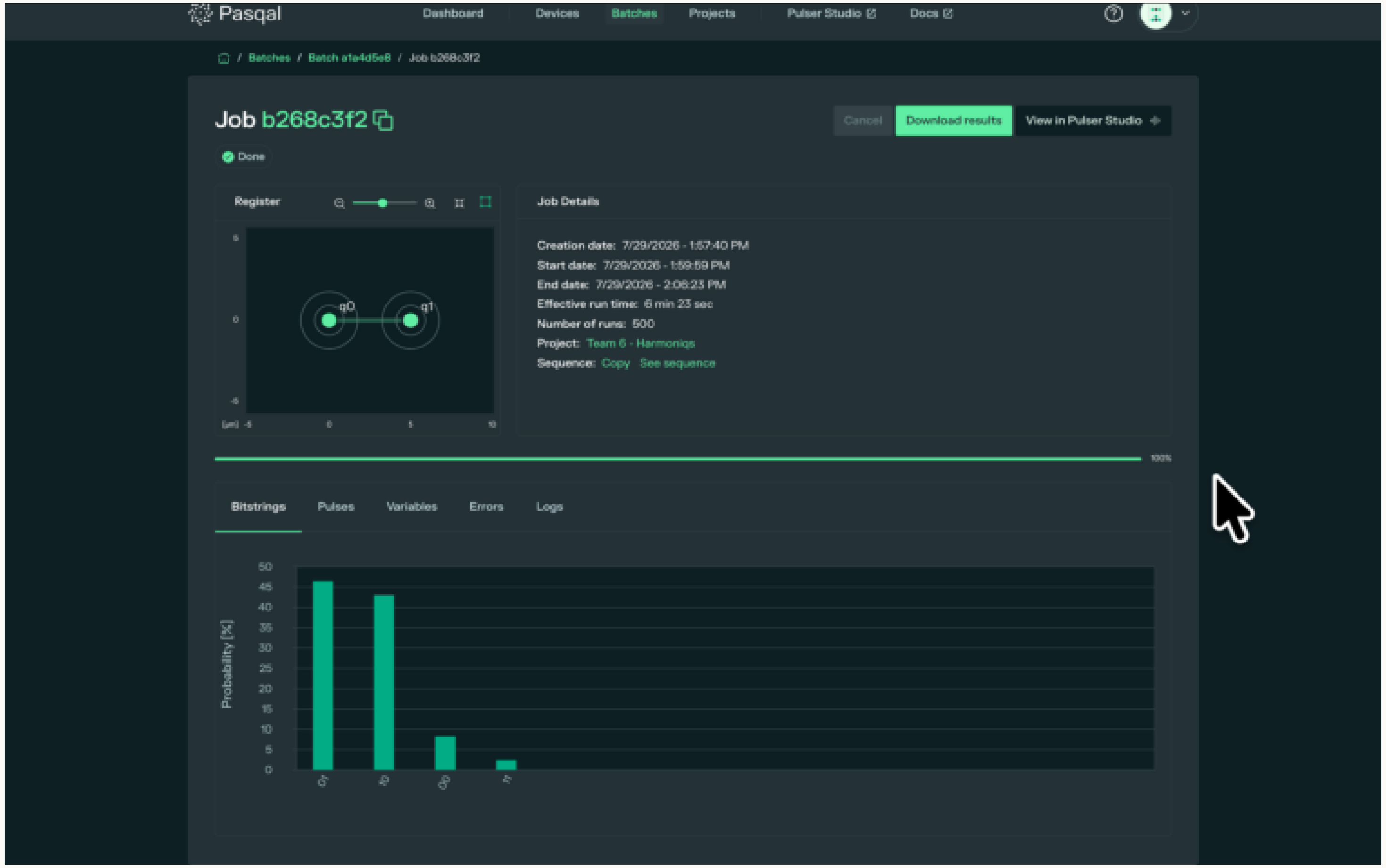
All scores from the same exact simulator the judges use; cloud columns are measured, 500 shots each.



top row: our pulse shapes (x: time; blue: power Ω, red: pitch δ) · bottom-left: score bars vs baseline · bottom-right: cloud measurements (dots) vs simulation (circles) — they agree

Grey bars are the starter pulse: 0.9926 and 0.7500. Green and blue bars are ours: 0.999+ everywhere.

On the real machine: 89.4% entangled



Pasqal's own results page · x: the four possible photos of two atoms · y: % of 500 shots

- We sent our 260-nanosecond pulse to FRESNEL_CAN1, in France.
- The two tall bars are the Bell state: one atom ON, either one — together 89.4% of all 500 photos.
- The bars are equal, as the physics demands. The forbidden both-ON result appears in only 2.4%.
- We published the expected range before the run — the result landed inside it.

Same laser, bigger question: can the atoms compute?

- Challenge 1 treated the blockade as an obstacle. Challenge 2 uses it as a resource:
- The no-two-ON rule is exactly the rule of a famous puzzle — and the hardware enforces it for free.

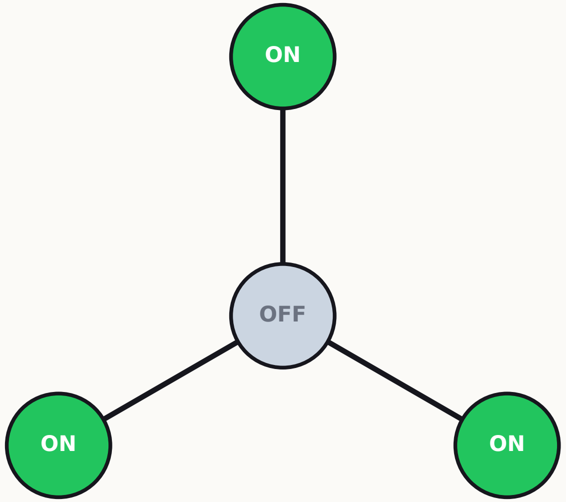
Also carried forward from Challenge 1:

- Short pulses beat long ones on real hardware (noise grows with time).
- One referee program per challenge; predictions published before every hardware run.

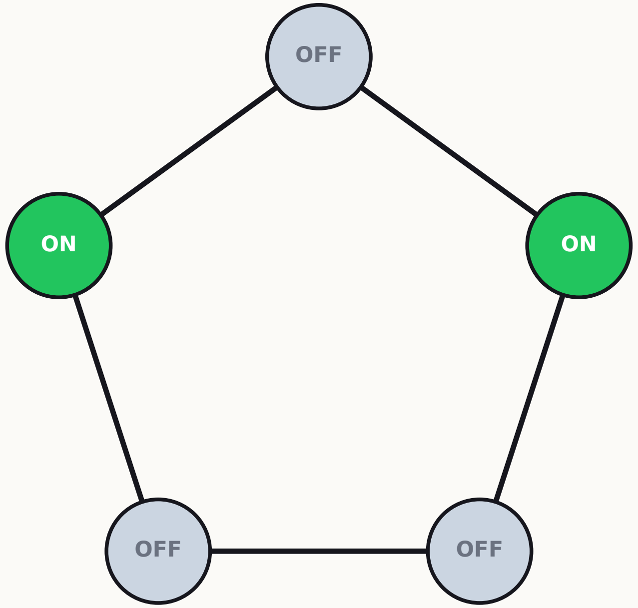
Challenge 2 — the seating puzzle

Maximum Independent Set (MIS): seat the most people so that no two sit next to each other.

- GIVEN: these two graphs, and a starter sweep scoring 0.727 (star) and 0.657 (pentagon).
- WE design: where each atom sits (lines = pairs closer than R_b) and the sweep Ω(t), δ(t). A photo at the end reads out the answer: ON atoms are the chosen seats.



STAR graph (4 atoms): best answer is the 3 outer atoms ON — only one answer.



PENTAGON (5 atoms): any 2 non-neighbors ON — five equally correct answers.

How we solved it, step by step

1

Draw the graph with atoms

place atoms so 'closer than R_b' gives exactly the target edges — verified in code

→

2

Build the referee

P_MIS from the exact state; self-tests catch convention mistakes

→

3

Reproduce the baseline

0.727 and 0.657, from the brief's own sweep parameters

→

4

Optimize the sweep

GRAPE again — but aimed at ALL correct answers at once (the pentagon has 5)

→

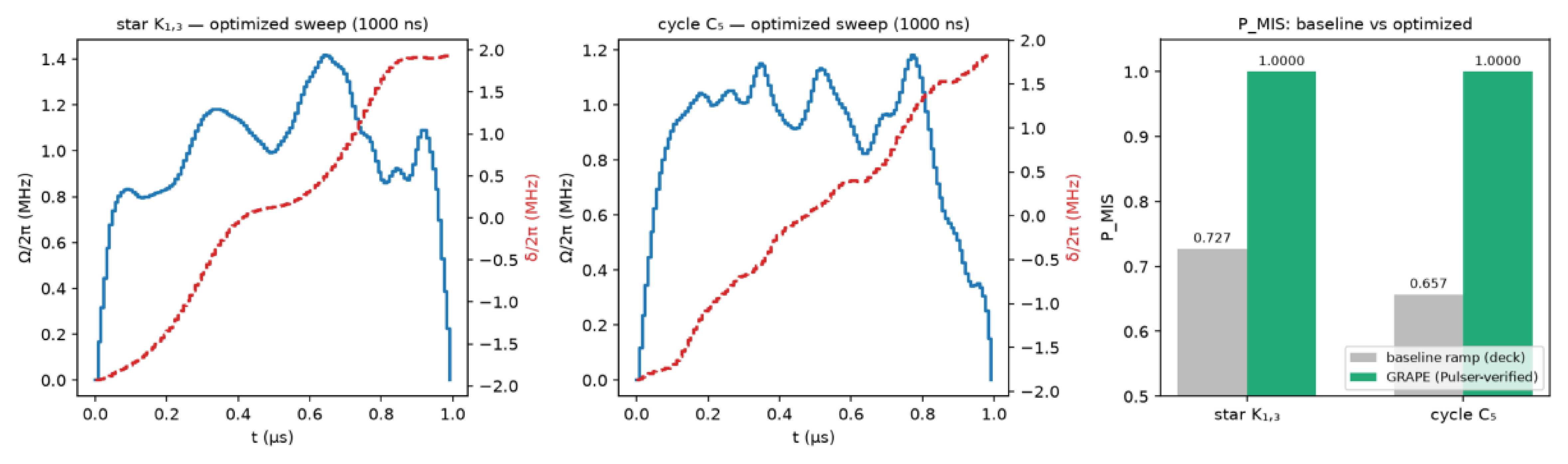
5

Simplify, then validate

a 5-knob version for hardware → cloud → the real machine

- The key idea in step 4: the starter sweep is slow and careful, yet still spills. We stop being careful and let the optimizer find the best route directly — scores jump to 0.999998 and 0.999999.
- And our sweeps take 1000 nanoseconds — one quarter of the starter's 4000 — which wins big under noise.

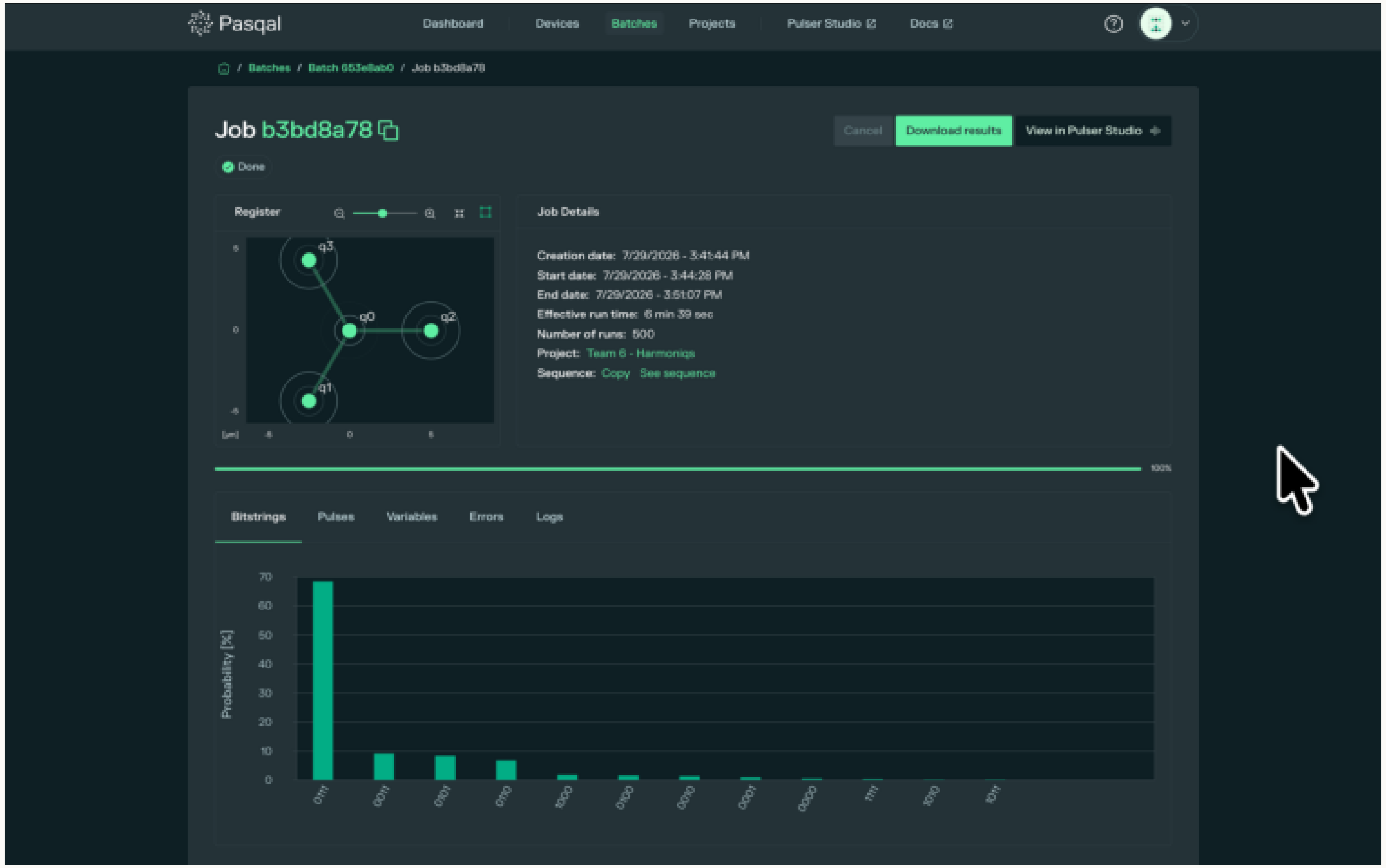
Challenge 2 results — and 1500 perfect photos



left/middle — x: time (μs), blue: power Ω, red: pitch δ · right — x: puzzle, y: score P_MIS

- On Pasqal's cloud: 1500 shots across three sweeps, and every single photo showed a correct answer.
- The pentagon's five answers came back 114, 98, 97, 96, 95 times — evenly, as quantum theory predicts: the machine holds all five answers at once and samples them fairly.

Real atoms pick the right answer, 7 to 1



Pasqal's results page · top-left: our star, drawn by the machine itself · bottom — x: measured pattern, y: % of 500 shots

- The register view shows the machine holding OUR graph: hub plus three leaves, in real optical tweezers.
- The towering bar, 0111, is the correct answer: 68.4% of 500 photos.
  The runner-up gets 9.2%.
- That is below the 80% we predicted: keeping three atoms ON compounds readout losses. We published the prediction first, the analysis after.

The score sheet

Challenge 1 — fidelity F (score defined in the brief)

	close pair (5.0 µm)	far pair (6.5 µm)	pulse length
Starter pulse	0.9926	0.7500	352 ns
Ours (best)	**1.000000**	**1.000000**	**224–600 ns**
Real machine	89.4% entangled (500 shots)	n/a on this lattice	260 ns

Challenge 2 — answer probability P_MIS (score defined in the brief)

	star (4 atoms)	pentagon (5 atoms)	sweep length
Starter sweep	0.727	0.657	4000 ns
Ours (best)	**0.999998**	**0.999999**	**1000–2000 ns**
Cloud, 500-shot runs	500/500 correct	500/500, twice	—
Real machine	68.4% exact (7× runner-up)	n/a on this lattice	1000 ns

Both stages completed. Every number regenerates from our public repository; every hardware claim carries the machine's own batch ID.

What we're taking home

- Diagnose before you optimize: one physics ratio (push ÷ power) explained both baseline failures.
- On real hardware, time is the enemy: our pulses are 4–11× shorter than the starters, and that is exactly why they survive noise.
- Five simple knobs got within half a percent of full optimal control — and carried over to bigger puzzles without re-tuning. That is the recipe for the 80-atom instances of Challenge 3.
- Trust is a workflow: one referee per challenge, predictions published before both hardware runs, and the one miss analyzed in the open.
- Thank you to the Pasqal stack — the Pulser toolkit, the free cloud emulator, and the FRESNEL machine that drew our graph back at us.